5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

1.1.2 Data Representation

Information today comes in different forms such as text, numbers, images, audio, and video.

Text

In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a **code,** and the process of representing symbols is called coding. Today, the prevalent coding system is called **Unicode,** which uses 32 bits to represent a symbol or character used in any language in the world. The **American Standard Code for Information Interchange (ASCII),** developed some decades ago in the United States, now constitutes the first 127 characters in Unicode and is also referred to as **Basic Latin**.

Numbers

Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations. Appendix B discusses several different numbering systems.

Images

Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the *resolution.* For example, an image can be divided into 1000 pixels or 10,000 pixels. In the second case, there is a better representation of the image (better resolution), but more memory is needed to store the image.

After an image is divided into pixels, each pixel is assigned a bit pattern. The size and the value of the pattern depend on the image. For an image made of only black-and-white dots (e.g., a chessboard), a 1-bit pattern is enough to represent a pixel. If an image is not made of pure white and pure black pixels, we can increase the size of the bit pattern to include gray scale. For example, to show four levels of gray scale, we can use 2-bit patterns. A black pixel can be represented by 00, a dark gray pixel by 01, a light gray pixel by 10, and a white pixel by 11.

There are several methods to represent color images. One method is called **RGB,** so called because each color is made of a combination of three primary colors: *r*ed, *g*reen, and *b*lue. The intensity of each color is measured, and a bit pattern is assigned to it. Another method is called YCM, in which a color is made of a combination of three other primary colors: yellow, cyan, and magenta.

Audio

Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

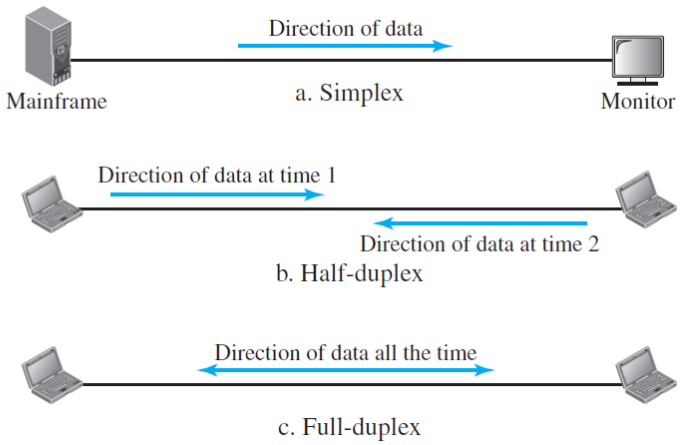
Video

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

1.1.3 Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in Figure 1.2.

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



Simplex

In **simplex mode,** the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see Figure 1.2a).

Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

Half-Duplex

In **half-duplex mode,** each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (see Figure 1.2b).

The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait. In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

Full-Duplex

In **full-duplex mode** (also called *duplex*), both stations can transmit and receive simultaneously (see Figure 1.2c).

The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction. This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.

CHAPTER 1: Computer Networks and the Internet

1.1 INTRODUCTION

The Internet is the largest engineered system ever built, connecting hundreds of millions of computers, communication links, and switches, as well as billions of users through devices such as laptops, smartphones, and an increasing variety of "smart" objects (IoT) like cars, watches, and home appliances. Despite its size and complexity, the Internet can be understood through guiding principles and structured concepts, making the study of computer networks both practical and engaging.

This chapter provides an overview of computer networking, beginning with fundamental terminology and concepts. It introduces the Internet's edge (end systems and applications) and its core (links, switches, access networks, and physical media), showing how the Internet is essentially a network of networks.

The chapter also discusses key performance aspects of networks, such as delay, loss, and throughput, and introduces quantitative models that account for transmission, propagation, and queuing delays. It presents fundamental networking principles like protocol layering and service models, while also highlighting security challenges and common types of attacks. Finally, it includes a brief history of computer networking.

When defining the Internet, two perspectives are used:

- Nuts-and-Bolts View – The Internet consists of billions of interconnected devices (hosts/end systems) such as computers, smartphones, tablets, and IoT devices.

- Service View – The Internet provides networking infrastructure that enables distributed applications like web browsing, email, and video streaming.

With billions of devices already online (18 billion in 2017, projected to reach 28.5 billion by 2022), the Internet continues to evolve as a global platform for communication, computation, and services.

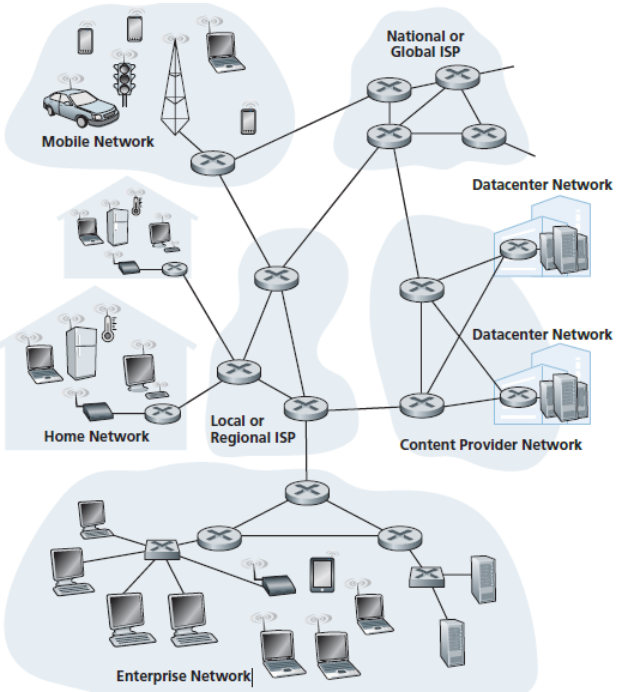
1.2 What Is the Internet?

1.2.1 A Nuts-and-Bolts Description

The Internet is a worldwide network of connected computers and devices. These devices include personal computers, smartphones, servers, and many more. The Internet connects them through communication links (cables, radio waves, satellites, etc.) and switching devices (like routers). Data is transmitted in small units called packets. Routers and switches direct these packets to their destinations.
The Internet Service Providers (ISPs) form the backbone of the Internet. They interconnect with each other to create a global network.

Detailed Explanation for Students:
Think of the Internet as a "network of networks." Your laptop at home connects to your ISP through Wi-Fi, DSL, cable, or fiber. Your ISP then connects to larger networks, and eventually, your data travels across multiple ISPs until it reaches the destination (for example, Google's data center).

- ✓ End systems (hosts): laptops, phones, servers.
- ✓ Communication links: fiber optic cables, copper wires, Wi-Fi, satellite.
- ✓ Packet switches: routers, switches, which forward data.
- ✓ Protocols: rules (like TCP/IP) that govern communication between devices.



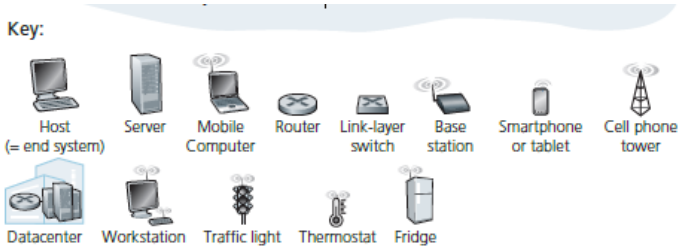


Figure 1.1 ♦ Some pieces of the Internet

The figure shows many computers, servers, and mobile devices connected through routers and communication links. It highlights how packets travel across the Internet backbone.

1.1.1 Internet Structure and Services

1. **End Systems and Data Transmission**
   End systems (e.g., PCs, smartphones, servers) are connected via communication links made of coaxial cable, copper, optical fiber, or wireless spectrum. Data is segmented into packets with headers and transmitted across the network, then reassembled at the destination.

2. **Packet Switching and Devices**
   Packets travel through a series of packet switches (mainly routers and link-layer switches).

✓ Routers: operate in the network core.

✓ Switches: used in access networks.
The sequence of links and switches forms the packet's route or path.

3. **Analogy with Transportation**
   Packets are compared to trucks transporting cargo on roads:

✓ Packets ↔ trucks

✓ Links ↔ roads/highways

✓ Switches ↔ intersections

✓ End systems ↔ buildings

4. **Internet Service Providers (ISPs)**
   ISPs connect end systems to the Internet (residential, corporate, mobile, etc.) and interconnect at different tiers. Upper-tier ISPs use high-speed routers and fiber links, while lower-tier ISPs provide local access. Each ISP runs IP and manages its own network independently.

5. **Protocols and Standards**
   The Internet relies on TCP/IP protocols:

✓ IP defines packet format.

✓ TCP provides reliable communication.
Standards are set by organizations such as IETF (publishes RFCs) and IEEE 802 (defines Ethernet and WiFi standards).

6. **The Internet as a Service Infrastructure**
   The Internet provides services to distributed applications (e.g., email, web, streaming, social media, video conferencing). Applications run on end systems, not in the network core. Communication is done through the socket interface, which specifies rules for data exchange between programs.

7. **Services Provided**
   Like postal services offering different delivery options, the Internet provides multiple service types (e.g., reliable TCP service vs. faster but less reliable UDP). Developers must select the appropriate service for their applications.

1.1.2 What Is a Protocol?

1. Human Analogy of Protocols

Humans follow protocols in communication without realizing it.

Example 1: Asking for the time.

Start with a greeting "Hi."

Wait for a response ("Hi" back → continue, "Don't bother me!" → stop).

Based on the reply (or no reply), you decide the next action.

Example 2: Classroom Q&A.

Teacher asks: "Any questions?"

Student raises hand → teacher acknowledges → student asks question → teacher answers.

Key point:
Communication works only if both parties follow the same set of rules. Otherwise, no useful communication occurs.

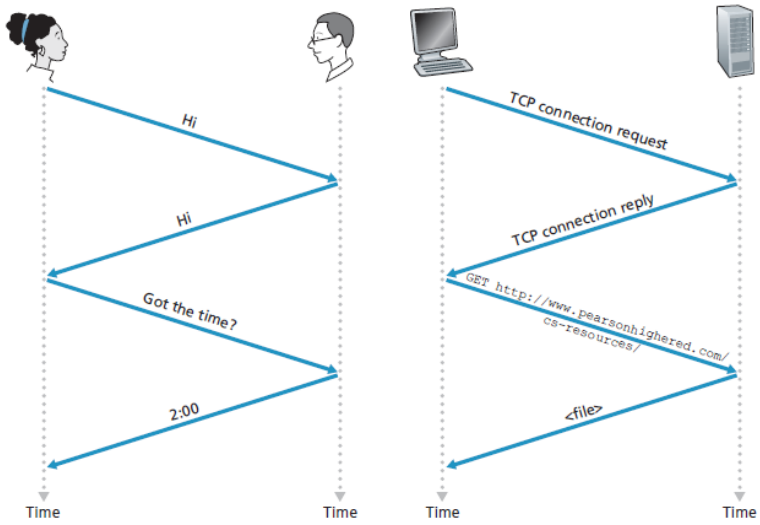


Figure 1.2 ♦ A human protocol and a computer network protocol

2. Network Protocols

In computer networks, protocols play the same role as human manners but between devices and software.

Examples:

- Protocols in network cards control the flow of bits over a physical wire.
- Congestion-control protocols manage data transmission rate between sender and receiver.
- Router protocols decide the path of packets across the network.
- Protocols run everywhere in the Internet—they are the foundation of all communication.

3. Web Request Example (HTTP protocol)

When you enter a URL in your browser:

1. Client → Server: Send a connection request.
2. Server → Client: Reply confirming the connection.
3. Client → Server: Send a GET request with the webpage name.
4. Server → Client: Send back the webpage (file).

This is a structured sequence of messages and responses, governed by the HTTP protocol.

4. Formal Definition of a Protocol

A protocol specifies:

1) Format of messages (structure, headers, content).

2) Order of message exchange between entities.

3) Actions taken when messages are transmitted, received, or when specific events occur.

5. Importance

- The Internet relies on a wide variety of protocols, each designed for different communication tasks.
- Some are simple, others are complex and deep.
- Mastering networking = understanding the design, role, and operation of these protocols.

1.2 – The Network Edge

1. From general to specific:

After introducing the Internet and protocols at a high level, this section focuses on the network edge — the part of the Internet where users directly interact with devices.

2. Definition of End Systems (Hosts):

- The devices connected to the Internet (PCs, Macs, Linux computers, laptops, smartphones, tablets, servers, and IoT devices).
- They are called end systems because they are located at the edge of the Internet.
- Also called hosts because they host applications like web browsers, email clients, web servers, etc.
- End system = Host.

3. Classification of Hosts:

- Clients: End-user devices such as desktops, laptops, tablets, and smartphones.
- Servers: More powerful machines that provide services (web pages, videos, email, cloud apps).
- Servers are usually located in large data centers.

✓ Example: As of 2020, Google operates 19 data centers across 4 continents, containing millions of servers.

4. Importance of the Network Edge:

- This is the starting and ending point of Internet communication.
- Applications and user interactions happen at this layer (e.g., browsing, video streaming, emailing).
- Data centers act as the backbone for delivering large-scale services and content to clients.

1.2.1 Access Networks

The access network is the physical connection that links an end system (like a home PC or smartphone) to the edge router of the Internet. Different types of access networks include:

✓ DSL (Digital Subscriber Line): Uses telephone lines for Internet.

✓ Cable Internet (HFC – Hybrid Fiber Coax): Uses the same cables as cable TV.

✓ FTTH (Fiber to the Home): Provides very high-speed Internet via optical fiber.

✓ 5G Fixed Wireless: Uses wireless technology instead of cables to deliver broadband.

✓ Ethernet: Common in enterprises, universities, and homes (wired LAN).

✓ Wi-Fi (802.11): Wireless access for local areas (home, offices, cafes, airports).

✓ 3G/4G/5G cellular: Mobile Internet through cellular towers.

Access networks are like the "on-ramps" of the Internet.

- At home, you might connect through DSL, cable, or fiber.
- At a university, you connect via Ethernet or Wi-Fi.
- On the move, your smartphone uses 4G or 5G.

Each access method has advantages and limitations:

- DSL and cable depend on existing infrastructure.
- FTTH gives extremely high speed but is expensive.

- Wi-Fi is flexible but limited in range.
- 5G wireless is powerful for mobility.

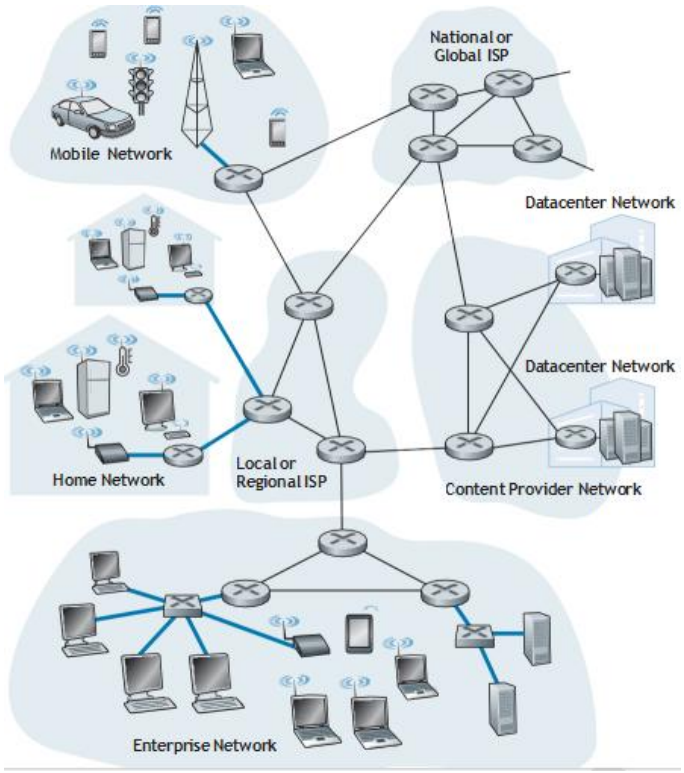


Figure 1.3 Access networks

Shows different access networks (home with DSL/cable, enterprise with Ethernet, and wide-area with 4G/5G).

1.2.2 Physical Media

When data travels across the Internet, it moves as bits through a physical medium. Physical media are of two types:

- Guided media (wired): The signal is guided along a solid path.

✓ Twisted-pair copper wire: Cheap, common in LANs (10 Mbps–10 Gbps).

✓ Coaxial cable: Used in cable TV and Internet (high-speed shared medium).

✓ Fiber optics: Very high bandwidth, long distances, immune to interference (used in backbones, FTTH).

- Unguided media (wireless): The signal travels freely through air/space.

✓ Terrestrial radio channels: Wi-Fi, Bluetooth, cellular.

✓ Satellite radio channels: Geostationary or Low-Earth Orbit satellites for global coverage.

Think of media as the roads that carry Internet "traffic." Some are copper highways, some are fiber expressways, and some are wireless air routes. Each medium has different speed, cost, and limitations.

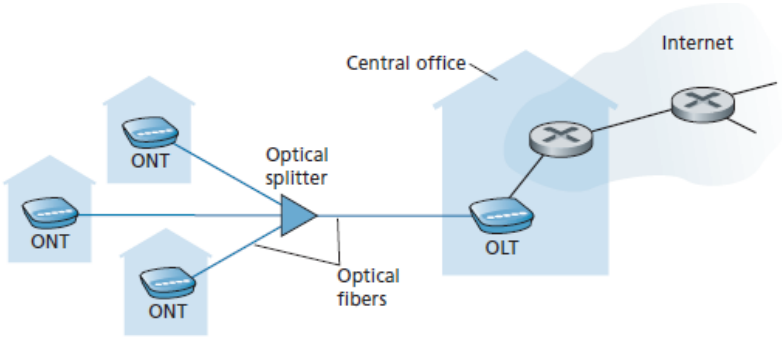


Figure 1.4 FTTH Internet access

Figure 1.4 shows FTTH using PON architecture (splitters, optical network terminator, OLT).

1.Cable Internet (Hybrid Fiber Coax - HFC)

Uses cable TV infrastructure.

Cable modem ↔ CMTS (Cable Modem Termination System).

Shared medium: bandwidth is divided among 500–5000 homes.

Asymmetric (download faster than upload).

Example rates: up to 1.2 Gbps downstream, 100 Mbps upstream (DOCSIS 3.0).

2.FTTH (Fiber to the Home)

Direct optical fiber connection from the ISP to the home.

Extremely high speeds (Gbps range).

Uses either:

AON (Active Optical Network) → like switched Ethernet.

PON (Passive Optical Network) → splitter divides fiber for multiple homes, common in Verizon FiOS.

3.5G Fixed Wireless

- Wireless broadband from provider's base station to the home modem.
- No need for physical cabling.
- Uses beamforming to deliver high data rates.
- Home router then distributes via WiFi/Ethernet.

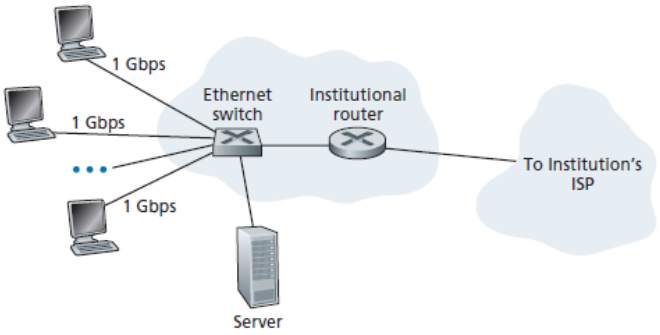


Figure 1.5 Ethernet Internet access

Figure 1.5 shows Ethernet and Wi-Fi access in enterprises and homes.

4. Enterprise and Home LAN Access

- Ethernet (Wired LAN):
- Uses twisted-pair copper wire.
- Connected to Ethernet switches.
- Speeds: 100 Mbps → 10 Gbps+.
- WiFi (Wireless LAN - IEEE 802.11):
- Uses access points connected to Ethernet backbone.
- Coverage: tens of meters.
- Speeds: 100 Mbps+ (shared).
- Home Networks often combine:
- DSL/Cable/FTTH access + WiFi routers.

Supports multiple devices (PCs, smartphones, IoT).

5. Wide-Area Wireless Access

- 3G, 4G LTE, and 5G mobile technologies.
- Connect smartphones/laptops directly to base stations operated by telecom companies.
- Coverage: several kilometers (much larger than WiFi).

Speeds:

✓ 4G → ~60 Mbps real-world.

✓ 5G → even higher (Gbps potential).

1.3 The Network Core

The network core is the central part of the Internet. It is made of packet switches (routers, switches) and communication links that interconnect millions of end systems.

If the network edge is like houses and driveways, then the network core is like the highways and intersections that connect all cities together.

1.3.1 Packet Switching

- Packet Switching is a method used by the Internet to send data efficiently.

Instead of sending an entire message as one big block, the message is divided into small pieces called packets. Each packet travels through the network separately — it may even take a different path — and all packets are reassembled at the destination.

- Each link in the network transmits data at a specific speed called the transmission rate (R bits/sec). If a packet has L bits, the time it takes to send it over one link is called the transmission delay, calculated as $d = L / R$.
- Routers in the network use a method called store-and-forward transmission, meaning each router must receive the entire packet before sending it to the next link. This adds delay at every step. So, if the message passes through N links, the total end-to-end transmission delay becomes $N \times (L / R)$.
- Each router also has a buffer (queue) where packets wait if the link is busy. When too many packets arrive at once, the waiting time increases — this is called queuing delay. If the buffer becomes full, new packets cannot be stored and are dropped (packet loss).

- Routers use forwarding tables to decide where each packet should go next, based on its destination IP address. These tables are built automatically using routing protocols such as shortest path algorithms.

Figure 1.6 shows the store-and-forward process, where a router waits to receive all bits of a packet before forwarding it.

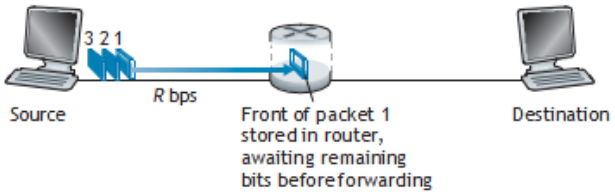


Figure 1.6 ♦ Store-and-forward packet switching

Figure 1.7 shows queuing at a router, where many packets arrive at the same time and must wait their turn to be sent.

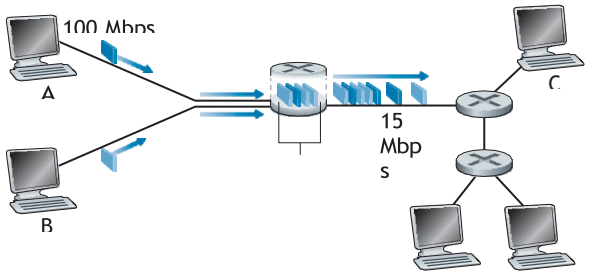


Figure 1.7 ♦ Packet switching

Fig 1.6 shows store-and-forward switching, where a router waits to receive all bits before forwarding.

Fig 1.7 shows packets queuing at a router when many hosts send data.

1.3.2 Circuit Switching

Circuit Switching is a communication method in which network resources (like bandwidth and buffer space) are reserved along the entire path between the sender and receiver before communication begins.

It's similar to making a reservation at a restaurant — once the reservation is made, your table (or in this case, the communication path) is guaranteed for you, even if you are not using it all the time.

Telephone Networks

Traditional telephone systems use circuit switching.

When a call starts, a dedicated communication circuit is established between the caller and the receiver, and it remains active for the entire duration of the call.

This means that no other users can use those reserved resources until the call ends.

Multiplexing Techniques

Circuit switching uses multiplexing to allow multiple users to share the same physical link efficiently. Two common methods are:

1. Frequency-Division Multiplexing (FDM)

   The link's total bandwidth is divided into separate frequency bands.

   Each call uses its own frequency band continuously.

   Example: Like different radio stations each using a unique frequency.

2. Time-Division Multiplexing (TDM)

   Time is divided into repeating frames with fixed time slots.

   Each call gets its own time slot to send data.

   Example: If a link has 1.536 Mbps and 24 slots, each slot provides 64 kbps.

   So, each voice call gets a guaranteed 64 kbps of bandwidth.

Problems of Circuit Switching

Although circuit switching provides guaranteed performance, it has several drawbacks:

Inefficient use of resources: When a user is silent (for example, during pauses in conversation), the reserved bandwidth is wasted.

Complex setup: Establishing the dedicated path requires signaling and coordination between network devices.

Not suitable for bursty data traffic: Internet applications like web browsing or video streaming send data in bursts, not continuously, making circuit switching inefficient for them.

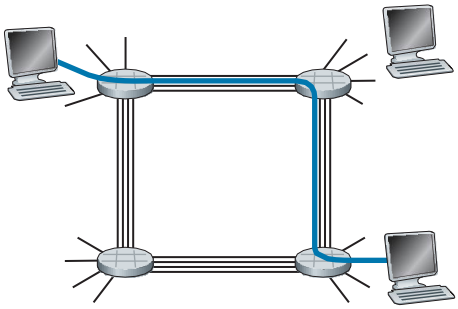


Figure 1.8 ♦ A simple circuit-switched network consisting of four switches and four links

Figure 1.8 : Shows a simple circuit-switched network with four switches connected by links. Each call sets up a dedicated path through these switches.

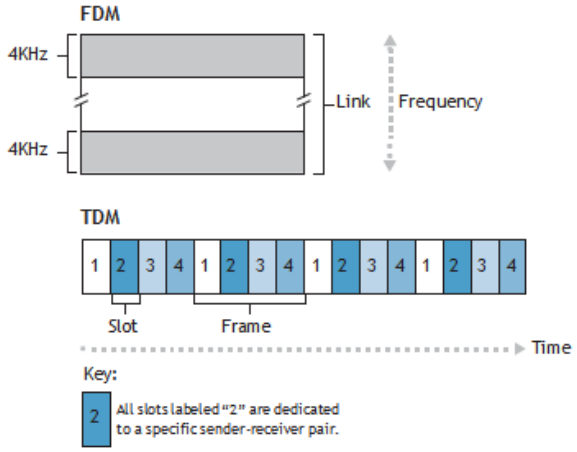


Figure 1.9 ♦ With FDM, each circuit continuously gets a fraction of the bandwidth. With TDM, each circuit gets all of the bandwidth periodically during brief intervals of time (that is, during slots)

Figure 1.9: Illustrates FDM and TDM multiplexing, showing how multiple calls share the same physical link using either different frequency bands (FDM) or different time slots (TDM).

1.3.3 A Network of Networks

1. What is the idea?

The Internet is not one single network, but rather a collection of many networks (ISPs) that are connected together.

That's why we call it a network of networks.

Each smaller network — like the network of your university, your home provider, or a company — connects to the Internet through an access ISP (Internet Service Provider).

2. Access ISPs and Interconnection

- End devices like laptops, phones, and servers connect to an access ISP.
- Access ISPs can use different technologies (DSL, fiber, Wi-Fi, 4G/5G, etc.).
- But just connecting people to their access ISP isn't enough — these ISPs themselves must also connect to each other.

That's how data can flow globally.

3. Why not just connect every ISP to every other?

That would require hundreds of thousands of connections, which is too expensive and complex.

So, instead, the Internet is built in layers or tiers of ISPs.

Network Structures (Evolution of the Internet)

Network Structure 1

- All access ISPs connect to one global transit ISP (a huge worldwide network).
- This global ISP provides international connectivity.
- Each access ISP pays this global provider (customer–provider relationship).

Network Structure 2

- There are multiple competing global transit ISPs.
- Access ISPs can choose which global provider to use.
- These global ISPs must also interconnect with one another so traffic can flow between them.

This forms a two-tier hierarchy:

Top: Global (Tier-1) ISPs

Bottom: Access ISPs

Network Structure 3

- Introduces Regional ISPs between access and global ISPs.

✓ Example: Local access ISPs → Regional ISP → Tier-1 ISP

In some countries (like China), there may even be provincial and national levels.

Each lower tier pays the higher tier for connectivity.

Tier-1 ISPs are at the top — they don't pay anyone because they already have full global connectivity.

Network Structure 4

Adds new components that make the Internet closer to reality:

✓ PoPs (Points of Presence): router groups where customers connect into a provider's network.

✓ Multi-homing: connecting to more than one provider for redundancy.

✓ Peering: two ISPs connect directly to exchange traffic without paying each other (settlement-free).

✓ IXPs (Internet Exchange Points): facilities where many ISPs meet and peer together.

There are over 600 IXPs globally.

So, in this structure, you have:

> Access ISPs → Regional ISPs → Tier-1 ISPs + Peering + IXPs + Multi-homing.

Network Structure 5 (Today's Internet)

Now we add content-provider networks, like Google, Meta, Amazon, etc.

✓ These companies build their own global private networks with huge data centers.

✓ They peer directly with lower-tier ISPs (to reduce cost and improve speed).

✓ But they still connect to Tier-1 ISPs for global reach.

Example:

Google's network connects its 19+ global data centers, bypasses upper tiers where possible, and peers directly with ISPs through IXPs.

Summary of the Modern Internet

- A multi-layered hierarchy:

✓ Tier-1 ISPs (about a dozen global ones)

✓ Regional and national ISPs

✓ Access ISPs

✓ Content provider networks

- The relationships are mostly customer–provider (money paid for service) and peering (mutual, free exchange).
- It's complex and partly shaped by economics and policy, not just by technology.

1.4 — Delay, Loss, and Throughput

Once we understand the Internet's structure, we study how data moves inside it.

Delays: time a packet takes to travel from source to destination (includes processing, queuing, transmission, propagation).

Loss: packets may be dropped if routers' buffers overflow.

Throughput: how much data per second can actually be transferred.

Even though we wish data could move instantly and perfectly, real networks introduce latency, congestion, and limits — and studying these gives insight into how the Internet actually performs.

Would you like me to continue with Section 1.4 (Delay, Loss, and Throughput) next — also in English explanation form with diagrams and examples (like the toll-booth analogy)?

This passage explains several key networking concepts — packet loss, delay, throughput, and protocol layering — in simple and structured English.

Here's a detailed explanation and summary for each main part:

1. Packet Loss

In theory, we often assume that a router's queue (the place where packets wait before being sent) can hold an unlimited number of packets.

But in reality, queues have limited capacity — once they are full, new packets are dropped (lost) because there's no more space.

- **Why does it happen?**

When the traffic intensity (how busy the network is) becomes too high — greater than what the router can handle — packets arrive faster than they can be processed or transmitted, causing overflow.

- **From the user's view:**

It looks as if the packet "disappeared" — sent but never arrived at the destination.

- **Impact:**

Packet loss increases when the network is busy, so network performance is usually measured by delay and packet loss rate.

Lost packets might be retransmitted later to ensure all data eventually arrives correctly.

2. End-to-End Delay

This is the total delay from the source to the destination, passing through multiple routers.



$$d_{end-to-end} = N(d_{proc} + d_{trans} + d_{prop})$$

Where:

- d_{proc}: Processing delay (time to check headers and errors)
- $d_{trans} = \frac{L}{R}$: Transmission delay (time to push all bits of the packet into the link)
- d_{prop}: Propagation delay (time for bits to travel through the link)

In real networks, queuing delays and heterogeneous link speeds can make this more complex.

3. Traceroute

Traceroute is a tool that helps us measure end-to-end delay and see the path packets take across routers.

- It sends a sequence of special packets, each marked with a number (1, 2, 3, …, N).
- Each router that receives a packet sends back a small reply message containing its name and address.

- By recording the time between sending and receiving replies, Traceroute can show:

✓ Each router's address and name

✓ The round-trip delay (time to go there and back)

If a router doesn't reply (perhaps due to packet loss), Traceroute shows an asterisk ().

Example:

In the example given, packets traveled through 14 routers from the U.S. (UMass) to France (Sorbonne University).

Round-trip times varied, especially across the transatlantic link, which caused large delays due to long physical distance.

4. End-System and Application Delays

Delays don't just occur inside routers — they can also happen in end systems (computers):

In shared networks (e.g., Wi-Fi), a device might wait before sending to avoid collisions.

In VoIP (Voice over IP), there's packetization delay — time needed to fill a packet with voice data before sending it.

This delay affects the quality of voice calls.

5. Throughput

Throughput is the rate (in bits per second) at which data is successfully transferred from source to destination.

Instantaneous throughput: how fast data is being received right now.

Average throughput: total bits received divided by total time.

$$\text{Average throughput} = \frac{F}{T} \text{ bits/sec}$$

where F = file size, T = time to transfer it.

Examples:

If a file of 32 million bits is transferred over a 1 Mbps link, it takes about 32 seconds.

In a network with multiple links, throughput is limited by the bottleneck link:

$$\text{Throughput} = \min(R_1, R_2, ..., R_N)$$

where R_i are the link speeds.

Shared link example:

If 10 clients share a 5 Mbps link, each gets only 0.5 Mbps, even if their access links are faster.

6. Protocol Layers and Service Models

The Internet is very complex, so engineers organize it into layers, each responsible for specific tasks.

This is known as the layered architecture or protocol stack.

Human analogy – Airline system:

Just like flying involves several steps (ticketing, baggage, gate, takeoff, routing, landing), networking involves multiple layers.

Each layer:

- ✓ Performs its own function, and
- ✓ Uses the services of the layer below it.

This modular design allows easy maintenance — we can update one layer's implementation without changing others.

7. Internet Protocol Stack (Five Layers)

1. Application Layer – Where network apps and protocols work

Examples: HTTP (web), SMTP (email), FTP (file transfer), DNS (name translation).

Unit of data: Message

2. Transport Layer – Moves messages between applications

TCP: reliable, connection-oriented (used by web, email).

UDP: fast but unreliable, connectionless (used in video/audio streaming).

Unit of data: Segment

3. Network Layer – Moves packets (called datagrams) from one host to another.

Main protocol: IP (Internet Protocol)

Also includes routing protocols to choose paths.

Unit of data: Datagram

4. Link Layer – Moves frames between directly connected devices (e.g., Ethernet, Wi-Fi).

Unit of data: Frame

5. Physical Layer – Transmits raw bits over physical media (cables, radio waves).

In Summary:

Concept	Meaning
Packet Loss	Packets dropped due to full queues
End-to-End Delay	Total time from source to destination
Traceroute	Tool to measure path and round-trip delays
Throughput	Rate of successful data delivery
Layered Architecture	Organizes network functions into levels
Internet Protocol Stack	5 layers: Application, Transport, Network, Link, Physical

Summary

The Internet is a vast global network that interconnects millions of computers, mobile devices, and sensors, enabling communication and information exchange worldwide. It is built on a hierarchical structure of Internet Service Providers (ISPs) that connect users through routers and communication links such as fiber optics, copper cables, and wireless technologies. Data is transmitted in the form of packets following standardized protocols like TCP/IP, which define how information is divided, addressed, sent, and reassembled.

The core of the Internet uses packet switching, where data is broken into packets that travel independently through routers before being reassembled at the destination. This approach

efficiently shares network resources compared to circuit switching, which reserves a fixed path for each connection.

Every packet experiences different types of delay — processing, queuing, transmission, and propagation — and may face loss when router buffers overflow. The overall throughput of a connection depends on the slowest link along its path.

The Internet architecture follows a layered model (Application, Transport, Network, Link, and Physical layers), where each layer performs a specific function and interacts with the layer above and below it. The application layer supports user services like email and the web, while the transport layer (TCP or UDP) ensures reliable or fast data delivery.

In summary, Chapter 1 explains how the Internet works as an interconnected system of hardware, software, and protocols that together provide a scalable, flexible, and efficient platform for global communication and data exchange.

Theoretical Questions – Chapter 1

1. Explain in detail the main components of the Internet architecture. How do hosts, routers, communication links, and protocols work together to enable end-to-end communication?

2. Discuss the differences between circuit switching and packet switching. Which method is more efficient under modern Internet traffic patterns, and why?

3. Describe the types of delays that can occur in a packet-switched network. How do these delays contribute to end-to-end performance degradation?

4. Analyze the layered architecture of Internet protocols (TCP/IP model). Explain how modularity and encapsulation improve network design and troubleshooting.

5. Evaluate the concept of Internet Service Providers (ISPs) and the hierarchical structure of the Internet. How does this structure affect routing, performance, and global connectivity?

Chapter 1: Computer Networks and the Internet — MCQs

1. Which of the following best describes the Internet?

A. A single large network owned by one company

B. A collection of interconnected networks using standardized protocols

C. A satellite communication system

D. A wireless-only network

2. What is an end system in the Internet architecture?

A. A device that forwards packets only

B. A network router

C. A device that runs network applications like a PC or smartphone

D. A device that provides physical connectivity only

3. The Transmission Control Protocol (TCP) provides:

A. Best-effort delivery

B. Reliable, connection-oriented communication

C. Connectionless communication

D. Only error detection

4. The Internet Service Provider (ISP) hierarchy includes:

A. Only access ISPs

B. Access, regional, and tier-1 ISPs

C. Local routers only

D. Peer-to-peer users

5. Which of the following is not a form of network edge device?

A. Laptop computer

B. Smartphone

C. Router

D. Web server

6. In packet switching, long messages are divided into:

A. Frames

B. Segments

C. Packets

D. Signals

7. In store-and-forward transmission, a router must:

A. Start transmitting as soon as it receives the first bit

B. Receive the entire packet before forwarding

C. Compress packets before sending

D. Drop packets automatically

8. The time to transmit a packet of length L bits over a link of rate R bits/sec is:

A. R/L

B. L × R

C. L/R

D. R – L

9. If there are N links between source and destination, each of rate R, the total end-to-end delay (ignoring propagation) for one packet is:

A. L/R

B. N × L/R

C. L/(N × R)

D. (N – 1) × R

10. Queuing delay mainly depends on:

A. Propagation speed

B. Level of congestion and buffer size

C. Number of routers only

D. Link capacity only

11. Packet loss occurs when:

A. The router’s output buffer is full

B. Transmission rate equals packet size

C. A link becomes idle

D. A packet has no destination address

12. In the Internet, packet forwarding decisions are made based on:

A. Source IP address

B. Destination IP address

C. MAC address only

D. Port number

13. A forwarding table in a router is used to:

A. Store user data

B. Encrypt packets

C. Map destination addresses to output links

D. Compress headers

14. Routing protocols are used to:

A. Determine shortest paths and configure forwarding tables automatically

B. Monitor user traffic

C. Increase bandwidth

D. Manage physical links

15. In circuit switching, resources along the path are:

A. Shared dynamically among users

B. Reserved for the duration of the session

C. Used only when packets arrive

D. Always free for reuse

16. Frequency-Division Multiplexing (FDM) divides the link's:

A. Bandwidth into frequency bands

B. Time into frames

C. Signal into bits

D. IP addresses into ranges

17. Time-Division Multiplexing (TDM) divides the link's:

A. Frequency spectrum

B. Frames into frequency bands

C. Time into slots assigned to each connection

D. Queue into packets

18. Packet switching is generally more efficient than circuit switching because:

A. It uses reserved bandwidth

B. It allocates resources only when needed

C. It provides constant delay

D. It requires complex signaling

19. The main disadvantage of packet switching for real-time services is:

A. Packet loss and variable delay

B. Fixed allocation of bandwidth

C. Inability to share resources

D. High setup time

20. Which statement best summarizes the modern trend in communication networks?

A. Circuit switching is replacing packet switching

B. Packet switching is replacing circuit switching

C. Both are disappearing

D. Only telephone networks remain circuit-switched